



LINUX COMMANDS CHEATSHEET

**LAST MINUTE REVISION GUIDE FOR
INTERVIEWS**

1 - SYSTEM INFORMATION

<code>uname -a</code>	# Display Linux system information
<code>uname -r</code>	# Display kernel release information
<code>cat /etc/redhat-release</code>	# Show which version of Red Hat installed
<code>lsb_release -a</code>	# Show which version of Ubuntu installed
<code>uptime</code>	# Show how long the system has been running + load
<code>hostname</code>	# Show system host name
<code>hostname -I</code>	# Display all local IP addresses of the host
<code>last reboot</code>	# Show system reboot history
<code>date</code>	# Show the current date and time
<code>cal</code>	# Show this month's calendar
<code>w</code>	# Display who is online
<code>whoami</code>	# Who you are logged in as

2 - HARDWARE INFORMATION

<code>dmesg</code>	# Display messages in kernel ring buffer
<code>cat /proc/cpuinfo</code>	# Display CPU information
<code>cat /proc/meminfo</code>	# Display memory information
<code>free -h</code>	# Display free and used memory (-h for human readable, -m for MB, -g for GB.)
<code>lspci -tv</code>	# Display PCI devices
<code>lsusb -tv</code>	# Display USB devices
<code>dmidecode</code>	# Display DMI/SMBIOS (hardware info) from the BIOS
<code>hdparm -i /dev/sda</code>	# Show info about disk sda
<code>hdparm -tT /dev/sda</code>	# Perform a read speed test on disk sda
<code>badblocks -s /dev/sda</code>	# Test for unreadable blocks on disk sda
<code>lshw</code>	# Display information about CPU, memory, storage, and network interfaces

lsblk

Display information about all storage devices

3 - PERFORMANCE MONITORING AND STATISTICS

top

Display and manage the top processes

htop

Interactive process viewer (top alternative)

mpstat 1

Display processor related statistics

vmstat 1

Display virtual memory statistics

iostat 1

Display I/O statistics

tail -100 /var/log/messages

Display the last 100 syslog messages (Use /var/log/syslog for Debian based systems.)

tcpdump -i eth0

Capture and display all packets on interface eth0

tcpdump -i eth0 'port 80'

Monitor all traffic on port 80 (HTTP)

lsof

List all open files on the system

lsof -u user

List files opened by user

free -h

Display free and used memory (-h for human readable, -m for MB, -g for GB.)

watch df -h

Execute "df -h", showing periodic updates

mpstat

Display statistics about CPU usage

pidstat

Display statistics about processes running

4 - USER INFORMATION AND MANAGEMENT

id

Display the user and group ids of your current user.

last

Display the last users who have logged onto the system.

who

Show who is logged into the system.

w

Show who is logged in and what they are doing.

groupadd test

Create a group named "test".

useradd -c "John Smith" -m john

Create an account named john, with a

`userdel john`

comment of "John Smith" and create the user's home directory.

`usermod -aG sales john`

Delete the john account.

Add the john account to the sales group

5 - FILE AND DIRECTORY COMMANDS

`ls -al`

List all files in a long listing (detailed) format

`pwd`

Display the present working directory

`mkdir directory`

Create a directory

`rm file`

Remove (delete) file

`rm -r directory`

Remove the directory and its contents recursively

`rm -f file`

Force removal of file without prompting for confirmation

`rm -rf directory`

Forcefully remove directory recursively

`cp file1 file2`

Copy file1 to file2

`cp -r source_directory destination`

Copy source_directory recursively to destination. If destination exists, copy source_directory into destination, otherwise create destination with the contents of source_directory.

`mv file1 file2`

Rename or move file1 to file2. If file2 is an existing directory, move file1 into directory file2

`ln -s /path/to/file linkname`

Create symbolic link to linkname

`touch file`

Create an empty file or update the access and modification times of file.

`cat file`

View the contents of file

`less file`

Browse through a text file

`head file`

Display the first 10 lines of file

`tail file`

Display the last 10 lines of file

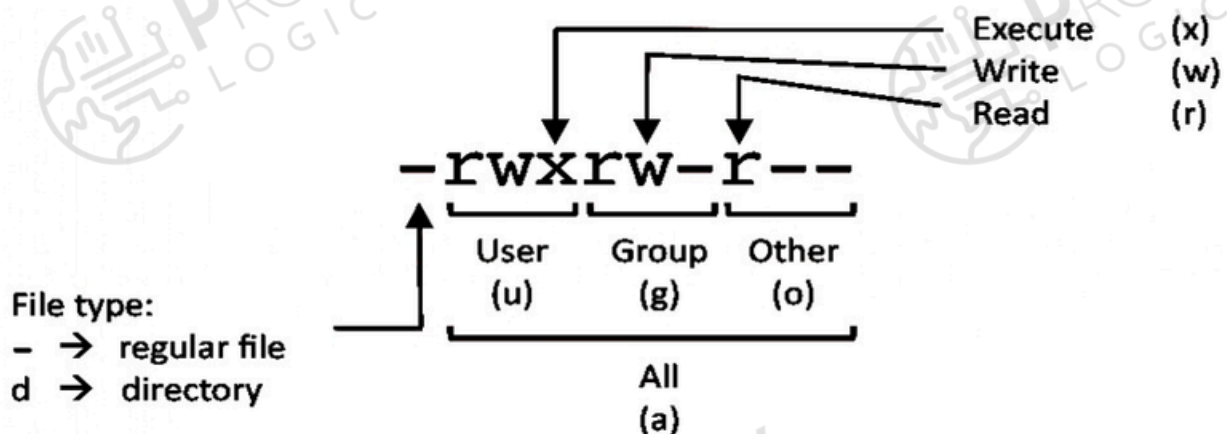
`tail -f file`

Display the last 10 lines of file and "follow" the file as it grows.

6 - PROCESS MANAGEMENT

<code>ps</code>	# Display your currently running processes
<code>ps -ef</code>	# Display all the currently running processes on the system.
<code>ps -ef grep processname</code>	# Display process information for processname
<code>top</code>	# Display and manage the top processes
<code>htop</code>	# Interactive process viewer (top alternative)
<code>kill pid</code>	# Kill process with process ID of pid
<code>killall processname</code>	# Kill all processes named processname
<code>program &</code>	# Start program in the background
<code>bg</code>	# Display stopped or background jobs
<code>fg</code>	# Brings the most recent background job to foreground
<code>fg n</code>	# Brings job n to the foreground
<code>nohup processname</code>	# Runs a process even after user logs out

7 - FILE PERMISSIONS



PERMISSION

U	G	W
<code>rwX</code>	<code>rwX</code>	<code>rwX</code>
<code>rwX</code>	<code>rwX</code>	<code>r-X</code>
<code>rwX</code>	<code>r-X</code>	<code>r-X</code>
<code>rw-</code>	<code>rw-</code>	<code>r--</code>

EXAMPLE

```

chmod 777 filename # Use sparingly!
chmod 775 filename
chmod 755 filename
chmod 664 filename
  
```

rw- r-- r-- chmod 644 filename

LEGEND

U = User

G = Group

W = World

r = Read

w = write

x = execute

- = no access

chown john /path/to/file

Change ownership of /path/to/file to john

chgrp sales /path/to/file

Change group ownership of /path/to/file to group sales

8 - NETWORKING

ip a

Display all network interfaces and IP address

ip addr show dev eth0

Display eth0 address and details

ethtool eth0

Query or control network driver and hardware settings

ping host

Send ICMP echo request to host

whois domain

Display whois information for domain

dig domain

Display DNS information for domain

dig -x IP_ADDRESS

Reverse lookup of IP_ADDRESS

host domain

Display DNS IP address for domain

hostname -i

Display the network address of the host name.

hostname -I

Display all local IP addresses of the host.

wget http://domain.com/file

Download http://domain.com/file

netstat -nutlp

Display listening tcp and udp ports and corresponding programs

ifconfig	# Display information about network interfaces
tracert host	# Display the path that packets take to host
tcpdump	# Capture and analyze network traffic

9 - ARCHIVES (TAR FILES)

tar cf archive.tar directory	# Create tar named archive.tar containing directory.
tar xf archive.tar	# Extract the contents from archive.tar.
tar czf archive.tar.gz directory	# Create a gzip compressed tar file name archive.tar.gz.
tar xzf archive.tar.gz	# Extract a gzip compressed tar file.
tar cjf archive.tar.bz2 directory	# Create a tar file with bzip2 compression
tar xjf archive.tar.bz2	# Extract a bzip2 compressed tar file.

10 - INSTALLING PACKAGES

yum search keyword	# Search for a package by keyword.
yum install package	# Install package.
yum info package	# Display description and summary information about package for RHEL based systems.
rpm -i package.rpm	# Install package from local file named package.rpm
yum remove package	# Remove/uninstall package for RHEL based systems.
yum update package	# Update package with name package for RHEL based systems.
tar zxvf sourcecode.tar.gz	# Install software from source code.
cd sourcecode	
./configure	
make	
make install	

<code>apt-get update</code>	# Update package list for Debian based systems.
<code>apt-get upgrade</code>	# Upgrade all installed packages to their newest version for Debian based systems.
<code>apt-get install package</code>	# Install package with name package for Debian based systems.
<code>apt-remove package</code>	# Remove package with name package for Debian based systems.

11 - SEARCH

<code>grep pattern file</code>	# Search for pattern in file
<code>grep -r pattern directory</code>	# Search recursively for pattern in directory
<code>locate name</code>	# Find files and directories by name
<code>find /home/john -name 'prefix*'</code>	# Find files in /home/john that start with "prefix".
<code>find /home -size +100M</code>	# Find files larger than 100MB in /home
<code>whereis program</code>	# Display the location of the binary, source and manual page files of program.
<code>which program</code>	# Display the path of executable that would run if program is executed.

12 - SSH LOGINS

<code>ssh host</code>	# Connect to host as your local username.
<code>ssh user@host</code>	# Connect to host as user
<code>ssh -p port user@host</code>	# Connect to host using port
<code>ssh-keygen</code>	# Create a new SSH key pair.
<code>ssh-copy-id user@host</code>	# Copy SSH key to the remote host to enable passwordless logins for user.

13 - FILE TRANSFERS

<code>scp file.txt server:/tmp</code>	# Secure copy file.txt to the /tmp folder on server
<code>scp server:/var/www/*.html /tmp</code>	# Copy *.html files from server to the local /tmp folder.
<code>scp -r server:/var/www /tmp</code>	# Copy all files and directories recursively from server to the current system's /tmp folder.
<code>rsync -a /home /backups/</code>	# Synchronize /home to /backups/home
<code>rsync -avz /home server:/backups/</code>	# Synchronize files/directories between the local and remote system with compression enabled
<code>ftp host</code>	# Connect to FTP server on the remote host.

14 - DISK USAGE

<code>df -h</code>	# Show free and used space on mounted filesystems
<code>df -i</code>	# Show free and used inodes on mounted filesystems
<code>fdisk -l</code>	# Display disks partitions sizes and types
<code>du -ah</code>	# Display disk usage for all files and directories in human readable format
<code>du -sh</code>	# Display total disk usage off the current directory
<code>du -a directory</code>	# Display size of all files in directory.
<code>findmnt</code>	# List all mounted file systems with details.

15 - DIRECTORY NAVIGATION

<code>cd ..</code>	# To go up one level of the directory tree. (Change into the parent directory.)
<code>cd</code>	# Go to the \$HOME directory
<code>cd /etc</code>	# Change to the /etc directory
<code>alias goto='cd /etc/'</code>	# Create goto alias for command cd /etc/.

16 - SECURITY

<code>passwd</code>	# Change the current user's password.
<code>sudo -i</code>	# Switch to the root account with root's environment. (Login shell.)
<code>sudo -s</code>	# Execute your current shell as root. (Non-login shell.)
<code>sudo -l</code>	# List sudo privileges for the current user.
<code>visudo</code>	# Edit the sudoers configuration file.
<code>getenforce</code>	# Display the current SELinux mode.
<code>sestatus</code>	# Display SELinux details such as the current SELinux mode, the configured mode, and the loaded policy.
<code>setenforce 0</code>	# Change the current SELinux mode to Permissive. (Does not survive a reboot.)
<code>setenforce 1</code>	# Change the current SELinux mode to Enforcing. (Does not survive a reboot.)
<code>SELINUX=enforcing</code>	# Set the SELinux mode to enforcing on boot by using this setting in the <code>/etc/selinux/config</code> file.
<code>SELINUX=permissive</code>	# Set the SELinux mode to permissive on boot by using this setting in the <code>/etc/selinux/config</code> file.
<code>SELINUX=disabled</code>	# Set the SELinux mode to disabled on boot by using this setting in the <code>/etc/selinux/config</code> file.

17 - LOGGING AND AUDITING

<code>dmesg</code>	# Display messages in kernel ring buffer.
<code>journalctl</code>	# Display logs stored in the systemd journal.
<code>journalctl -u servicename</code>	# Display logs for a specific unit (service).